



S. Sai Hemanth

Game Systems Engineer | Unity • XR •
High-Performance Systems

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Portfolio : [Technical Portfolio](#)

Professional Summary

Game systems engineer with 5+ years of experience building high-performance gameplay systems, XR applications, and real-time simulation tools in Unity (and Unreal). Specialized in data-oriented design, multi-threaded simulation, open-world streaming, event-driven architecture, and Meta Quest optimization. Strong focus on scalable, maintainable systems that perform under heavy load.

Core Skills

- **Game Engines** : Unity, Unreal
- **Programming Languages** : C#, C++, Python
- **Platforms** : PC, XR (Meta quest), Android, IOS
- **Specialties**: Gameplay systems, multi-threading (Burst/Jobs), rendering optimization, AI (GOAP), multiplayer, editor tools, data-oriented design

Professional Experience

Mad VR Solutions Pvt. Ltd.

Team Lead - R&D & Lead Unity and VR Developer | Remote | **June 2024 - August 2025**
(Company got restructured and removed entire dev team)

- Led the development of high-performance VR surgical simulators, improving real-time physics, interaction fidelity, and rendering efficiency.
- Designed and built scalable XR frameworks, enabling rapid prototyping for new simulators.
- Spearheaded performance optimization, ensuring Meta Quest store compliance and achieving 72+ FPS on standalone VR hardware.
- Developed VR hand tracking & physics-based interactions using Photon Fusion2 for multiplayer collaboration.
- Engineered custom shaders (HLSL) for advanced VR rendering, including screen-space effects, inverted hull outlines, and triplanar textures.
- Trained and mentored junior developers in VR fundamentals, rendering pipelines, and networking solutions.
- Provided technical guidance for integrating new VR hardware and motion-tracking systems.

One Immersive Pvt. Ltd.

Senior Unity Developer & XR R&D engineer | Hyderabad, India | Nov 2023 – June 2024

- Developed AR-based outdoor navigation using Mapbox VR for virtual real estate tours, enabling video streaming at landmarks.
- Built indoor AR navigation systems using Microsoft Spatial Anchors & Azure DB for precise indoor positioning.
- Designed a Geospatial AR Quest system, displaying AR banners and interactive content at significant city landmarks.
- Created a VR capture system similar to NVIDIA Ansel, using compute shaders to stitch stereo VR images inside Unity.

Turing – Crew Crafts Media

Unity Developer | Remote (California, USA) | Aug 2022 – Aug 2023

Project: Dance Kitchen (Retired app on Play Store) (Retired app on App Store)

- Led the development of Dance Kitchen, an interactive dance training app teaching Salsa, Bachata, and Mambo.
- Designed a dynamic content system using Timeline & Custom Data Events, enabling real-time updates via Digital Ocean.
- Engineered custom UI shaders for enhanced visuals and rendering performance.
- Built scalable footwork and body weight shift mechanics, allowing seamless instructor-led guidance in VR.
- Developed and optimized real-time audio synchronization with character animations for immersive dance lessons.
- Managed the full app development life-cycle, from UI programming to back-end integration and optimization.

IACG Multimedia

Unity Developer & Game Dev Professor | Hyderabad, Telangana | Oct 2021 – Dec 2023

- Developed custom shaders & rendering solutions, including a Triplanar Shader for far LOD objects, optimizing texture performance.
- Engineered IK systems using Unity's Animation Rigging, collaborating closely with the 3D team for character animation.
- Taught advanced game development skills to students, covering modular systems, Unity's lightmap switching, and optimization techniques.
- Mentored teams for Dancing Atoms' WCP Program, ensuring technical excellence in their projects.
- Worked on outsourced projects for Go Live Gaming, 5th Ocean Entertainment, and 7 Seas Entertainment.
- Introduced Utility AI & GOAP for complex NPC decision-making systems.

Selected Technical Projects

Leonids – High-performance data-driven ballistics (Unity + Unreal)

Data-oriented projectile system using NativeArray / parallel arrays, Burst/Jobs (or Unreal parallel processing), predictive collision, and zero runtime allocations. Supports up to 1M projectiles (Unity) / ~50k spawns/sec (Unreal). Event-driven feedback + GPU rendering (indirect / Niagara Data Channel).

Atlas – Open-world streaming framework (Unity)

Chunk-based (1000×1000 m) asynchronous streaming with Addressables. 3×3 active neighborhood, dynamic lighting support. Tested on full GTA: San Andreas map in HDRP at ~200 FPS on 4 GB VRAM.

Hermes – Event-driven gameplay messaging system (Unity)

Strongly-typed publish-subscribe bus that fully decouples systems. Used across personal and production XR projects.

Custom Real-Time Rendering Engine (C++ / HLSL)

Built from first principles: render graph, PBR, shadow mapping (PCF + cubemaps), asset importing, and skybox.

Other notable work: Janus (modular UE5 interaction framework), GOAP AI, Freedom Fighters (gameplay systems sandbox), volumetric black-hole renderer (SDF + ray marching).

Additional

- Independent Android game currently launching on itch.io
 - Workshops & tutoring (Unity game development)
 - Full technical portfolio & project deep-dives available on request / portfolio site
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If you have printed this resume please visit the link in the qr code below to see all my projects and works.

